AI Adoption, Productivity, and Growth: A Cross-Country Macroeconomic Analysis

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DATA 319 Final Project

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I. Overview

Artificial intelligence has become an important driver of economic activity, with growing effects on productivity, economic growth, and labor markets. However, countries differ widely in their ability to adopt and benefit from AI technologies. This project studies how AI adoption is related to macroeconomic performance across countries using a data-driven, multivariate approach.

Using international data on AI research, AI investment, AI skills, labor productivity, economic growth, and unemployment, this study applies correlation analysis, principal component analysis, clustering methods, and multivariate normality testing. The goal is to examine how AI adoption is linked to productivity and growth across countries and to highlight differences in national AI readiness, with implications for global economic performance and policy.

II. Background and motivation

Artificial intelligence has quickly become a major force shaping modern economies. Advances in machine learning, automation, and data processing have expanded the use of AI across many industries, including finance, healthcare, and public services. According to a 2024 report by the OECD on AI and productivity, AI is increasingly seen as a key driver of efficiency, innovation, and long-term economic growth.

Despite this strong potential, the benefits of AI are not shared evenly across countries. Advanced economies tend to lead in AI research, investment, and skills, while many developing economies fall behind due to weaker digital infrastructure, education systems, and access to capital. A 2025 OECD report on productivity gains from AI in G7 economies shows that early and intensive adopters of AI are already seeing faster productivity growth, raising concerns about widening global gaps.

Gaining the full benefits of AI requires more than just new technology. A 2025 working paper by the International Monetary Fund finds that countries with stronger human capital and institutions are better able to turn AI adoption into lasting economic growth.

Motivated by these global trends, this project studies how differences in AI adoption across countries are related to productivity and economic growth. Using cross-country data, the analysis aims to show whether countries with stronger AI readiness also perform better economically, and to highlight the key development and policy implications of AI.

III. Datasets selection and description

This project combines data from the OECD AI Data portal and the World Bank's World Development Indicators to capture both AI adoption and macroeconomic performance across countries. The datasets were selected to measure different dimensions of AI readiness and key economic outcomes relevant to productivity, growth, and labor markets.

III.1 AI Adoption and Readiness Datasets (OECD AI Data)

- AI Publications per Capita (ai_pubs_per_capita)
  This dataset measures the intensity of AI research activity relative to population size. It is used as an indicator of national AI innovation and knowledge production capacity.

- AI Publications over Time (ai_pubs_time)
  This dataset records the total volume of AI-related research output by country over time. It is used to examine broad trends in global AI research activity.

- AI Skills (ai_skills)
  This dataset measures the share of workers with AI-related skills based on LinkedIn data. It is used to capture the human capital dimension of AI readiness and to reflect workforce preparedness for AI adoption.

- AI Venture Capital Investment (ai_vc_investment)
  This dataset measures private-sector investment in AI by country. It is used as an indicator of AI commercialization and the strength of the AI innovation ecosystem.

III.2 Macroeconomic Outcome and Control Datasets (World Bank – World Development Indicators)

- Labor Productivity (labor_productivity)
  Measured as GDP per person employed, this dataset is used to capture economic efficiency and is a central outcome variable in evaluating the economic impact of AI.

- GDP Growth (gdp_growth)
  This dataset measures the annual real growth rate of GDP and is used to assess the relationship between AI adoption and overall economic development.

- Unemployment Rate (unemployment)
  This dataset measures labor market conditions and is used to examine whether AI adoption is related to changes in employment outcomes.

IV. Data challenges and preprocessing

Several challenges arose during data cleaning and merging due to differences in time coverage, country identifiers, and missing values across datasets. All sources were aligned using ISO country codes, and the analysis period was restricted to 2018–2023 to focus on recent AI expansion and maximize data overlap.

A key issue was the AI skills dataset, which is available only as a single country-level measure without yearly variation. This caused the variable to appear missing in the initial panel and was resolved by merging it at the country level and treating it as time-invariant.

Missing data were handled using method-specific complete-case filtering rather than imputation, which explains why sample sizes differed across methods. The multivariate normality test retained the largest sample because it relied on variables with the widest coverage.

Because AI variables differed in coverage, PCA was applied at the country level using average values to construct a balanced AI readiness index. Standard missing-value summaries were used, which count missing country–year observations rather than missing countries.

V. Research questions

What are the main factors that explain differences in AI adoption between countries?
This question is addressed by combining multiple AI indicators into a single AI readiness index using principal component analysis.

How does AI adoption relate to key economic indicators such as productivity and economic growth?
This question is examined using correlation analysis between AI-related variables and macroeconomic outcomes, including labor productivity and GDP growth.

Can countries be grouped into distinct clusters based on their AI readiness and economic performance?
This question is addressed using K-means clustering and hierarchical clustering techniques to identify groups of countries with similar AI–economic profiles.

Do the relationships among AI readiness, GDP growth, and productivity follow a multivariate normal distribution?
This question is examined using multivariate normality testing based on Mahalanobis distances and comparisons with simulated multivariate normal data.

VI. Methodology

This project uses multivariate statistical methods to study the relationship between AI adoption and macroeconomic performance across countries. The methods were chosen to fit the structure of the data, which contains several AI indicators and multiple economic outcome variables observed across countries and over time.

Among the possible methods considered, this study focuses on dimension reduction, clustering, correlation analysis, and multivariate normality testing.

Principal component analysis (PCA) is used to combine several AI indicators into a single AI readiness index. This makes it possible to summarize AI adoption with one measure instead of analyzing many highly correlated variables. The main benefit of PCA is that it simplifies the data

while keeping most of the original information. However, the resulting index is a statistical measure and does not have a direct economic unit.

Correlation analysis is used to examine the relationships between AI variables and key macroeconomic indicators such as productivity and economic growth. This method is simple and easy to interpret, but it only shows associations and does not imply causation.

Clustering methods, including K-means and hierarchical clustering, are used to group countries based on similarities in AI readiness and economic performance. These unsupervised learning methods help identify patterns and country groupings without pre-defined categories. A limitation is that the results can change depending on the number of clusters chosen and the variables included.

Multivariate normality testing is used to check whether the joint distribution of AI adoption, productivity, and growth follows standard statistical assumptions. This is important for assessing whether common multivariate models are appropriate. The analysis compares the real data with simulated multivariate normal data using Mahalanobis distances.

Other methods, such as supervised classification and formal hypothesis testing, were considered but not used because the project does not involve labeled outcome groups or a causal identification strategy. Instead, the selected methods focus on exploratory analysis, pattern discovery, and checking data distributions, which fit the goals of this study.

VII. Analysis

VII.1 Correlation matrix

Table 1 and the heatmap summarize the relationships between the main AI indicators and key macroeconomic variables, including labor productivity, GDP growth, and unemployment. This gives a first overview of how AI activity and economic performance are related across countries and over time.

A strong positive relationship is found between AI publications per capita and both labor productivity and GDP growth. This suggests that countries with more active AI research tend to

have higher productivity and faster economic growth. The relationship between labor productivity and GDP growth is also very strong, which matches standard economic expectations that more productive economies grow faster.

AI venture capital investment shows a positive but weaker correlation with productivity and growth. This implies that while private investment in AI is related to economic performance, its short-run link appears less direct than that of research activity. Differences in financial systems, innovation environments, and the timing of investment returns may explain this weaker relationship.

Unemployment has a negative relationship with both productivity and growth, meaning that stronger economic performance is usually linked to lower unemployment. The correlations between unemployment and AI-related variables are also negative but relatively small, suggesting that the labor market effects of AI are slower and more complex.

Overall, the correlation results provide early evidence that higher levels of AI activity, especially research, are linked to better economic performance. These results motivate the use of dimension reduction and clustering methods to further explore cross-country differences in AI readiness and macroeconomic outcomes.

	ai_pubs_per_capita	ai_vc_investment	labor_productivity	gdp_growth	unemployment
ai_pubs_per_capita	1.000000	0.116239	0.793686	0.839397	-0.172283
ai_vc_investment	0.116239	1.000000	0.151471	0.198566	-0.069296
labor_productivity	0.793686	0.151471	1.000000	0.918987	-0.130996
gdp_growth	0.839397	0.198566	0.918987	1.000000	-0.245221
unemployment	-0.172283	-0.069296	-0.130996	-0.245221	1.000000

Figure 1. Correlation matrix

Figure 2. Correlation matrix heatmap

VII.2 PCA on AI Variables

To combine different aspects of AI activity into one measure, principal component analysis (PCA) was applied to three country-level AI indicators: AI publications per capita, AI venture capital investment, and AI skills. The scree plot shows that the first principal component explains about 54% of the total variation, the second explains 34%, and the third explains the remaining 12%. This pattern indicates that one main factor captures most of the shared variation in AI activity across countries.

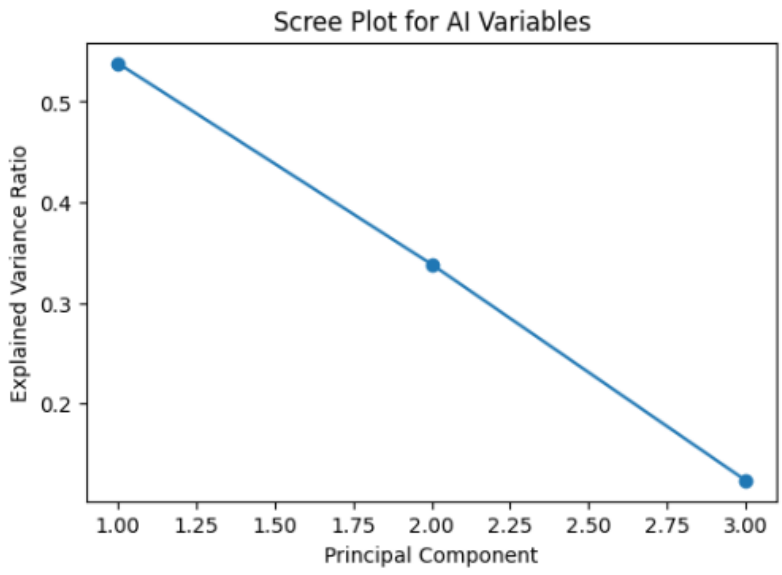


Figure 3. Scree plot for AI variables

The loading plot shows that all three AI variables load positively and with similar strength on the first principal component. This means that countries with high scores on this component tend to have strong AI research activity, high AI investment, and a larger share of workers with AI-related skills. For this reason, the first principal component is interpreted as an AI readiness

index that reflects a country's overall ability to develop and use AI.

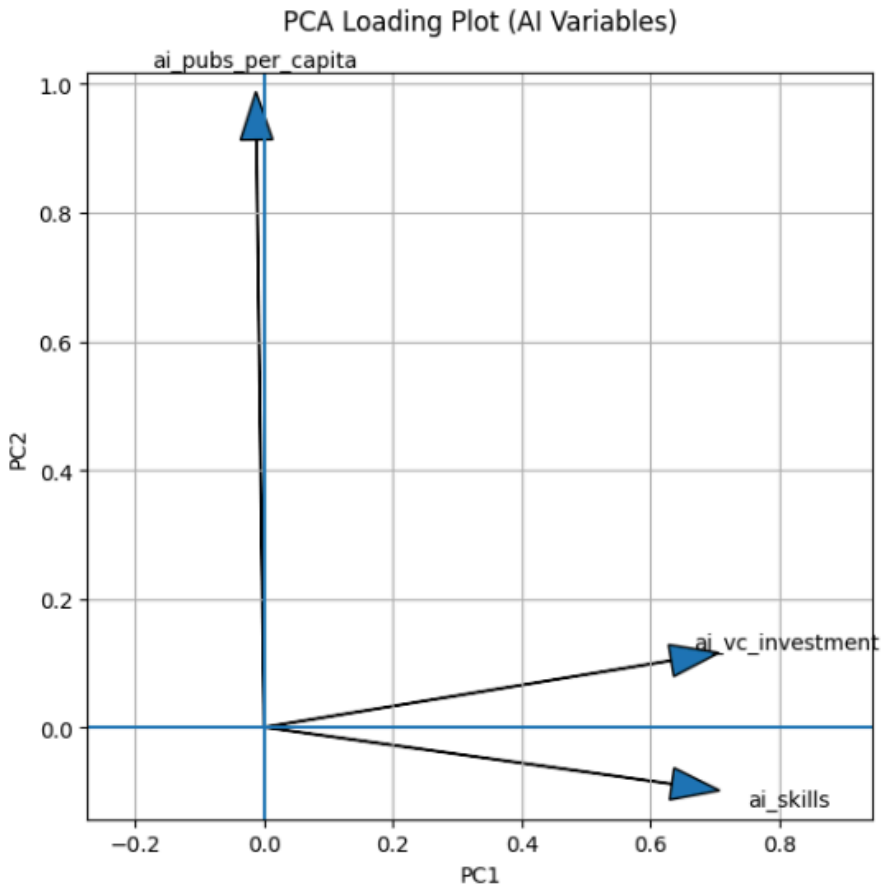


Figure 4. PCA loading plot (AI variables)

The PCA score plot shows how countries are positioned based on the first two principal components. Most countries are grouped near the center, indicating moderate levels of AI activity, while a small number stand out with much higher AI readiness. In particular, the United States appears far to the right on the first component, reflecting very high levels of AI publications and investment. Countries such as Singapore and Switzerland score highly on the second component, showing different patterns in AI skills and investment. These results highlight clear differences in AI readiness across countries and support the use of clustering

methods to group countries with similar AI–economic profiles.

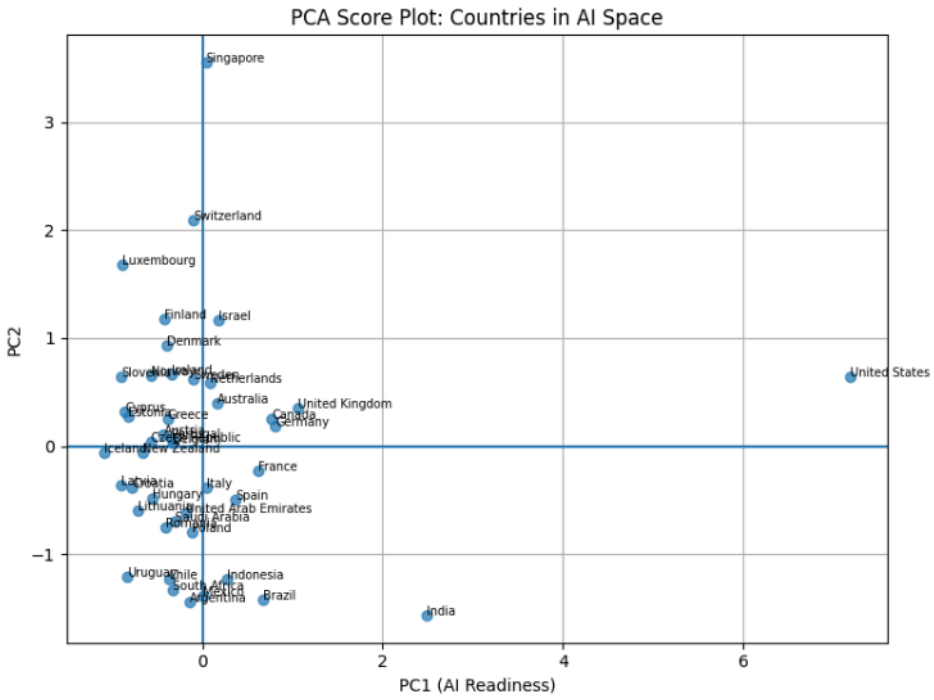


Figure 5. PCA Score Plot: Countries in AI Space

VII.3 K-means/Hierarchical Clustering

To further study differences in AI readiness and economic performance across countries, clustering methods were applied using the AI readiness index, labor productivity, and GDP growth. All variables were standardized before clustering so that differences in scale would not affect the distance-based results.

The elbow method was used to help choose the number of clusters. The elbow plot shows a steep drop in inertia from one to three clusters, after which the improvements become smaller. Based on this pattern, three clusters were chosen as a good balance between simplicity and explanatory power.

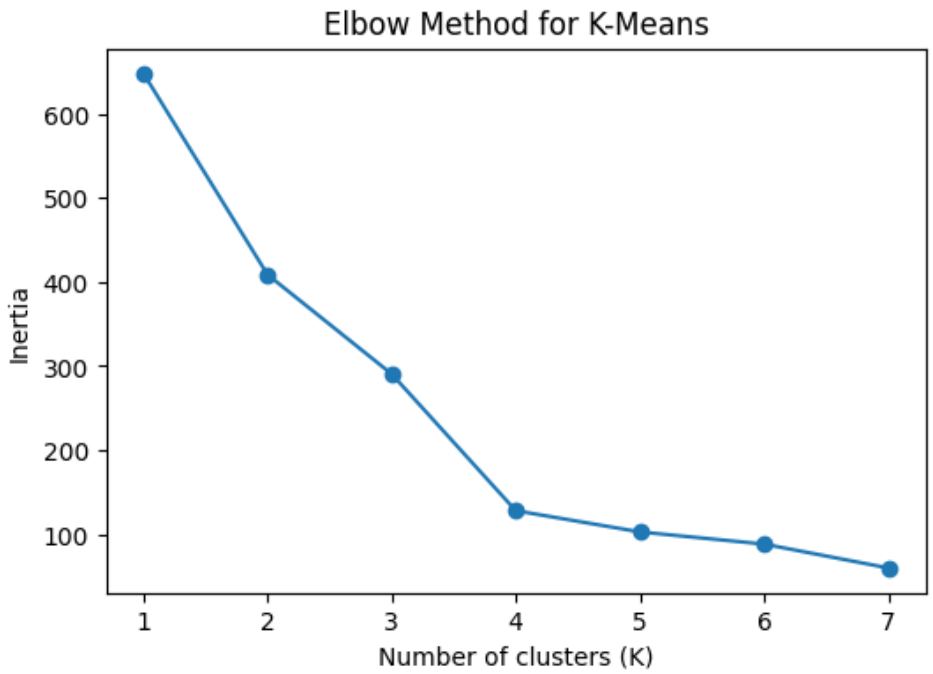


Figure 6. Elbow Method for K-Means

The K-means scatter plot of AI readiness versus labor productivity shows a clear separation among the three clusters. One cluster includes countries with high productivity and strong AI readiness, representing advanced economies that are well positioned to benefit from AI-driven growth. A second cluster contains countries with moderate productivity and mid-level AI readiness, reflecting economies that are actively adopting AI but have not yet reached the technological frontier. The third cluster is characterized by lower productivity and weaker AI

readiness, highlighting countries that are still at an early stage of AI adoption

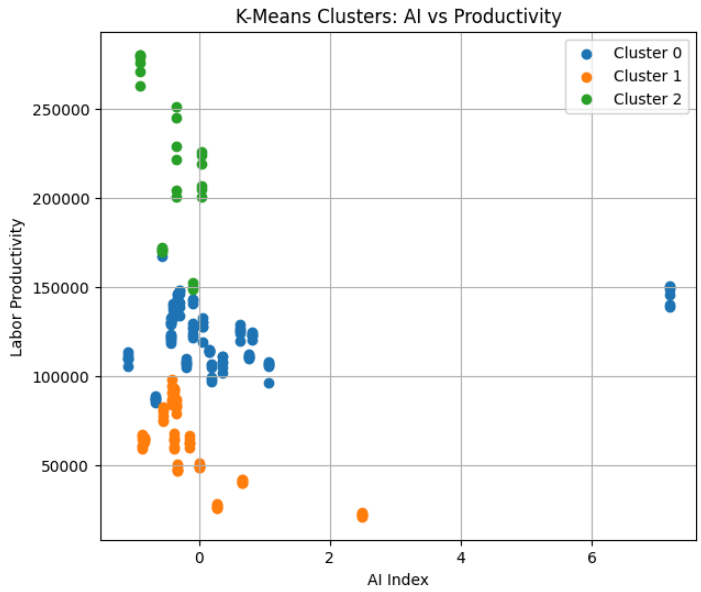


Figure 7. K-Means Clusters: AI vs. Productivity

These results suggest that AI readiness and productivity tend to move together across countries in a structured way, rather than being randomly related. Countries with stronger AI ecosystems also tend to show higher levels of economic efficiency.

Hierarchical clustering at the country level supports these findings. The dendrogram shows several clear country groupings based on similarities in AI readiness, productivity, and growth. Highly developed economies cluster closely together, while lower-income and emerging economies form separate groups. The agreement between the K-means and hierarchical

clustering results strengthens the reliability of these country groupings.

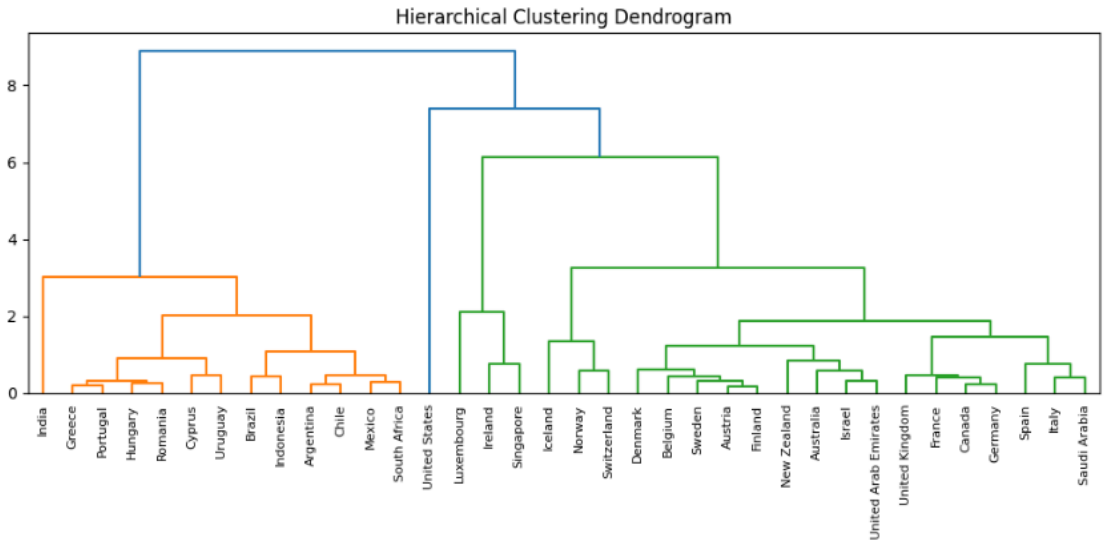


Figure 8. Hierarchical Clustering Dendogram

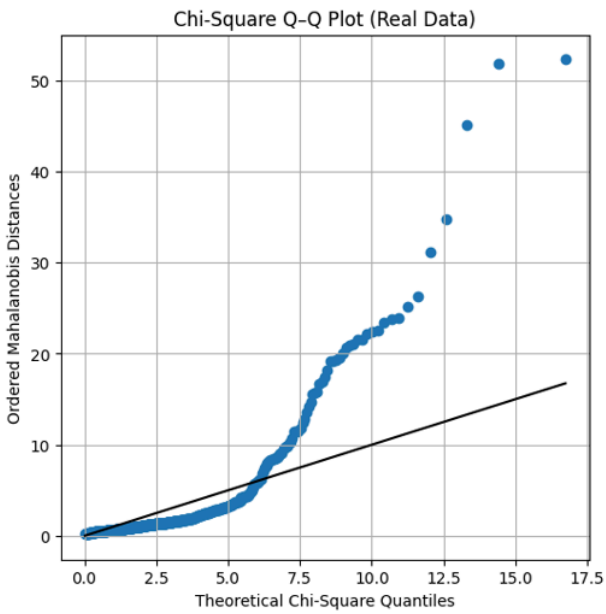
Overall, the clustering analysis shows clear global groupings in AI readiness and economic performance. These patterns suggest that AI adoption may be strengthening existing global economic gaps, with leading countries moving further ahead while others risk falling behind without targeted policy support.

VII.4 Multivariate Normality Testing

To check whether AI adoption, labor productivity, and GDP growth follow a joint multivariate normal distribution, a multivariate normality test based on Mahalanobis distances was applied. This test is important because many multivariate statistical methods assume that the data are jointly normal.

The Chi-square Q–Q plot for the real data shows clear departures from the 45-degree reference line, especially in the upper tail. While the lower part of the distribution follows the theoretical line fairly well, the largest observations lie far above it. This shows the presence of heavy tails and outliers in the joint distribution of AI research, productivity, and growth. In practical terms, this means that a small number of countries have unusually high levels of both AI activity and economic performance compared with the rest of the sample.

For comparison, data were simulated from a multivariate normal distribution using the same mean and covariance as the real data. The Q–Q plot for the simulated data closely follows the theoretical Chi-square line, as expected under normality. When the real and simulated results are shown together, the difference is clear: the simulated data match the theoretical pattern closely, while the real data show clear and systematic deviations.



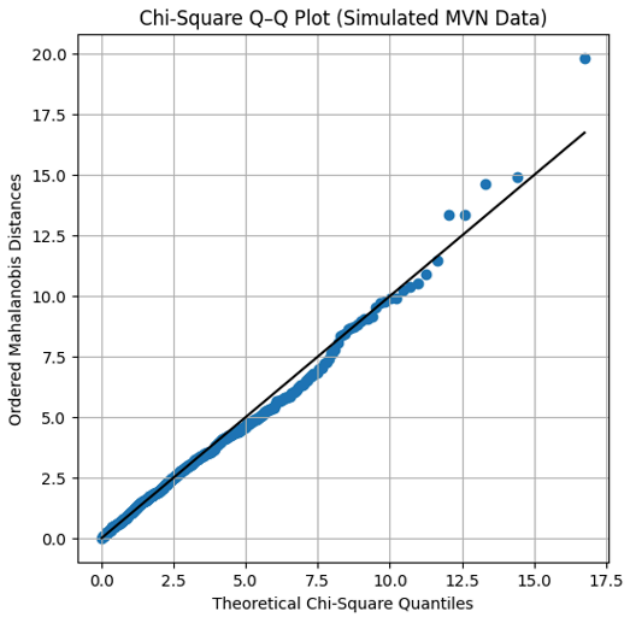


Figure 9. Chi-Square Q-Q Plot (Real Data)

Figure 10. Chi-Square Q-Q Plot (Simulated MVN Data)

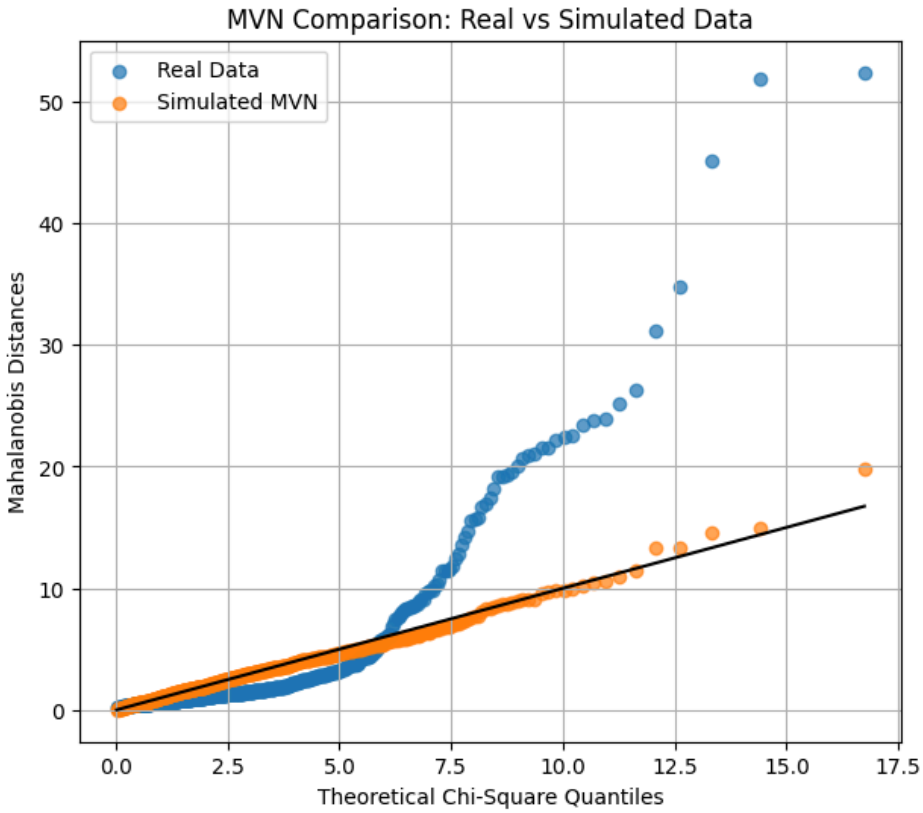


Figure 11. MVN Comparison: Real vs. Simulated Data

These results show that the joint distribution of AI adoption, productivity, and growth does not follow a multivariate normal pattern. This lack of normality is driven by large differences across countries and the presence of a few extreme performers. This means that caution is needed when using parametric multivariate models with this type of data, and it supports the use of exploratory, distribution-free methods such as correlation analysis, PCA, and clustering.

From an economic perspective, this departure from normality reflects the uneven global spread of AI. A small group of countries operates at a much higher level in terms of both AI activity and economic performance, reinforcing the idea that AI-driven growth is not evenly shared across the world.

VIII. Extensions and policy implications

While this project provides a broad cross-country view of the link between AI adoption and economic performance, several useful extensions would be possible with better data. One important extension would be a regional analysis of AI trends, comparing how AI adoption has developed across regions such as North America, Europe, East Asia, and emerging markets. This would help clarify regional technology gaps and the roles of institutions, trade links, and policy coordination in shaping AI development.

Another valuable extension would use more detailed time-series data on AI skills and AI investment. In this study, AI skills were treated as constant over time because yearly data were not available. With consistent annual data on workforce AI skills, it would be possible to study how changes in human capital directly affect productivity growth and labor market outcomes. Firm-level data on AI adoption would also help link innovation at the micro level to performance at the macro level.

Some countries were also excluded from parts of the analysis due to missing data, especially for AI investment and AI skills. As a result, many low-income and developing economies were underrepresented in the PCA and clustering results. Better global data coverage would allow future research to determine whether these countries are truly lagging in AI adoption or whether the gaps mainly reflect data limitations.

From a policy perspective, the results suggest that AI readiness, productivity, and economic growth are closely connected. Countries with stronger AI ecosystems tend to perform better economically. This highlights the importance of policies that support AI education and skills development, since human capital appears to be a key part of national AI readiness. Investment in digital infrastructure, data access, and research capacity is also likely to be crucial for wider participation in AI-driven growth.

For countries with lower AI readiness, targeted public investment, workforce retraining programs, and stronger support for innovation may help reduce the risk of falling further behind. At the international level, cooperation on AI standards, knowledge sharing, and capacity building

could play an important role in reducing global digital gaps and promoting more inclusive technological progress.

IX. Conclusion

This project studied how AI adoption is linked to productivity and economic growth across countries using multivariate statistical methods. The correlation analysis showed strong positive relationships between AI research activity, labor productivity, and GDP growth. The PCA-based AI readiness index also confirmed that AI research, investment, and skills together capture a common dimension of a country's overall AI capacity.

The clustering results revealed clear country groupings based on AI readiness and economic performance. High-income economies formed a distinct group with higher productivity and stronger AI ecosystems. The multivariate normality test showed that AI adoption, productivity, and growth do not follow a normal joint distribution, reflecting large differences across countries and the presence of a few extreme performers.

Overall, the results suggest that countries with stronger AI readiness tend to have higher productivity and faster economic growth. Although causality cannot be confirmed, the findings highlight the importance of investing in AI skills, innovation, and digital infrastructure to support long-run economic performance and help reduce global technology gaps.

XI. Presentation Feedback and Revisions

During the presentation, the professor pointed out that the original PCA score plot did not look realistic. Some countries that are well known for strong AI capabilities appeared too low, while others, like the United States, were extremely far to the right. This suggested that something in the data was pulling the PCA results in an unusual way.

Even though I had scaled the variables before running PCA, the AI variables themselves had very large differences in size. For example, AI venture capital investment in the United States is many times larger than in most other countries. These big gaps caused the PCA to be heavily influenced by a few extreme values.

To fix this, I applied a log-transformation to the three AI variables before scaling and running PCA again. The log-transform reduces the impact of countries with unusually high values by compressing the scale. After applying this step and updating the plots, the PCA results became much more reasonable. Countries now appear in positions that better match what we know from real-world AI readiness reports, and the relationships between countries are easier to understand.

The revised PCA score plot and loading plot (shown below) reflect these improvements and provide a clearer picture of global AI readiness.

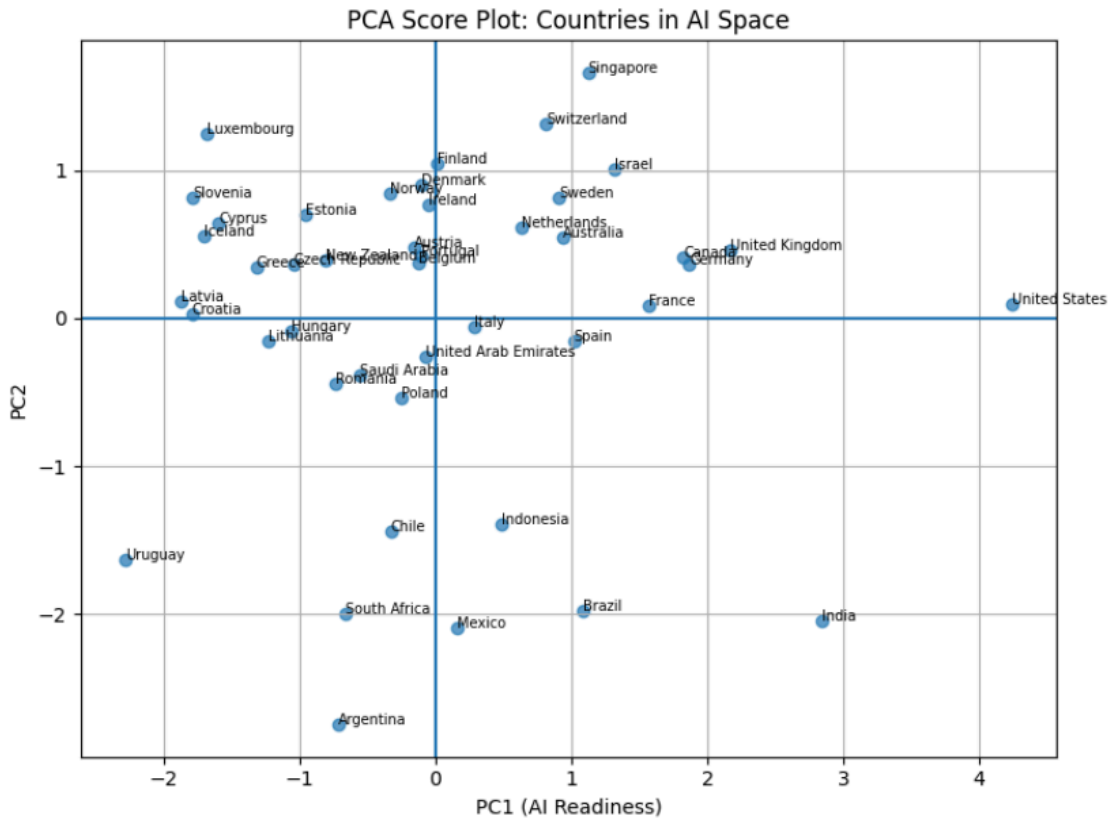


Figure 12. Revised PCA Score Plot

Figure 13. Revised K-Means Clusters: AI vs. Productivity

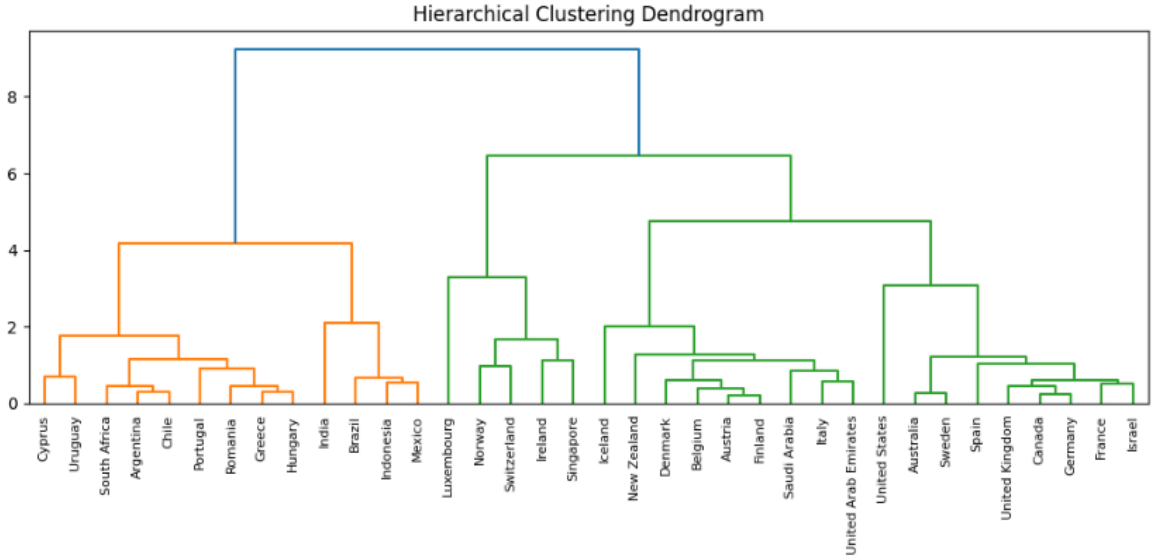


Figure 14. Revised Hierarchical Clustering Dendogram

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